

1) Duplicate check :-

Date

Input :- $a = \{4, 7, 2, 9\}$ (no duplicate, worst case)

Nested loops :-

$i=0; j=1, 2, 3 \rightarrow 4 \text{ vs } 7, 4 \text{ vs } 2, 4 \text{ vs } 9$	Comparison	3
$i=1; j=2, 3 \rightarrow 7 \text{ vs } 2, 7 \text{ vs } 9$		2
$i=2; j=3 \rightarrow 2 \text{ vs } 9$		1
$i=3; \text{ none}$		0

$$\text{Total} = 6 = n(n-1)/2 \rightarrow O(n^2)$$

return false

Hashset:

add(4) true, add(7) true, add(2) true, add(9) true $\rightarrow 4$ inserts $\rightarrow O(n)$ return false

With a duplicate, $\{4, 7, 4\}$, nested stops at $i=0, j=2$ because $4==4$. Hashset stops when add(4) returns false the second time.

Teacher's Signature

2) Java Stack Flow

Stack (Vector-based):

push A, B, C $\rightarrow$ [A, B, C]add(0, "SNEAK") $\rightarrow$ [SNEAK, A, B, C]

(inserted at bottom = break stack flow)

remove(2) $\rightarrow$ [SNEAK, A, C]get(1) $\rightarrow$ A

op 2 - java.util.Stack: [SNEAK, A, C]

Array Degree:

push A $\rightarrow$ [A]; push B $\rightarrow$ [B, A]; push C $\rightarrow$ [C, B, A]

pop() = C, then peek() = B

3) Linear Search Count

Input: $n=5$, $a = \{0, 3, 6, 9, 12\}$

Key	0	(index 0) $\rightarrow$ 1 comparison	best case
Key	6	(index 2) $\rightarrow$ 3 comparisons	
Key	12	(index 4) $\rightarrow$ 5 comparisons	middle case
Key	-7	(absent) $\rightarrow$ 5 comparisons	worst case
			worst case

Average: $(1+2+3+4+5)/5 = 3 = (n+1)/2$ $O(n)$

4) List Traps

ip: $n=4$, $[0, 1, 2, 3]$
`ArrayList.remove(0);`

remove $\rightarrow [1, 2, 3]$

shifted 3

remove $\rightarrow [2, 3]$

shifted 2

remove $\rightarrow [3]$

shifted 1

remove $\rightarrow []$

shifted 0

Total Shifts = $3+2+1+0 = 6 = n(n-1)/2 \rightarrow O(n^2)$

`ArrayDeque.poll()`: the head index just moves

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

That is 4 steps in total, so

$O(n)$ overall & $O(1)$ per poll.

5) StackAdd Demo

$\rightarrow$ reverse ("ABC");

push A $\rightarrow [A]$

push B $\rightarrow [A, B]$

push C $\rightarrow [A, B, C]$ ($\text{top} = c$)

Pop $\rightarrow c$, Pop $\rightarrow B$, Pop $\rightarrow A \rightarrow "CBA"$

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$$\begin{array}{r}
 26 \\
 - 13 \\
 \hline
 13 \\
 78 \\
 \hline
 238
 \end{array}$$

Date.....

6) String Concat Trap

$$n = 4$$

$S += 'x' \rightarrow$ new string of length 1 (copy 1)

$S += 'x' \rightarrow$ length 2 (copy 2)

$S += 'x' \rightarrow$ length 3 (copy 3)

$S += 'x' \rightarrow$ length 4 (copy 4)

$$\text{Total} = 1 + 2 + 3 + 4 = 10 \approx n(n+1)/2 \rightarrow O(n^2)$$

String Builder : n appends, with no re-copying. That is $O(n)$.

7) Program To Interface

Date _____

Input: $n=4$, $[0, 1, 2, 3]$

ArrayList: $\text{get}(i)$ jumps to the index,
No cost 1 step each & the loop is
 $O(n)$.

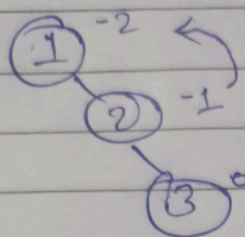
LinkedList: $\text{get}(i)$ walks node by node from
nearest end

RR - Left Rotation

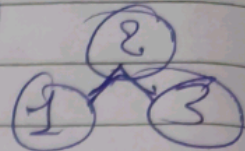
Date.....

Q) 1. 1, 2, 3, 4, 5, 6, 7

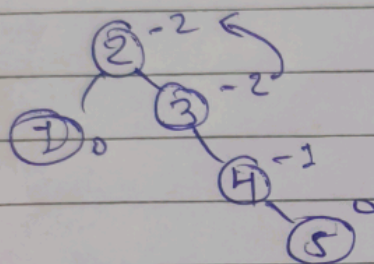
Q.1)



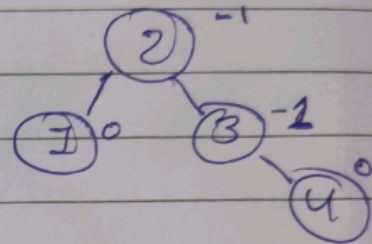
Left Rotation (1)
~~Right Rotation (1)~~



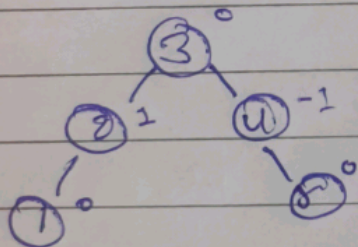
Insert 4



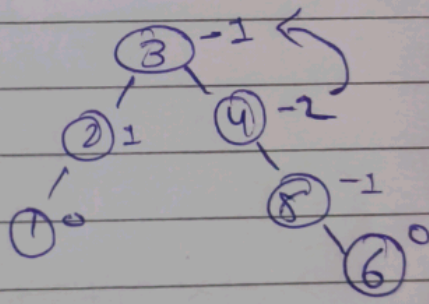
↔



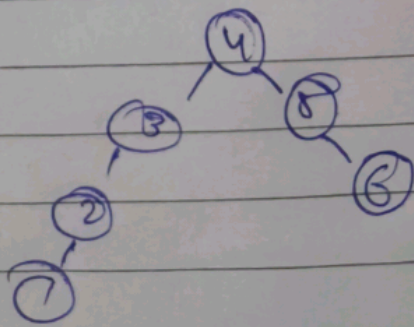
Left
Rotation
↙



→

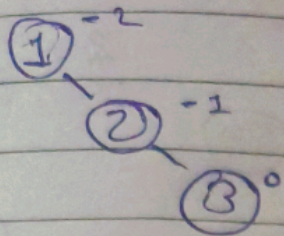


↘

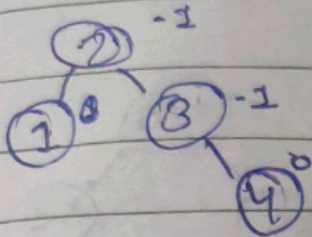
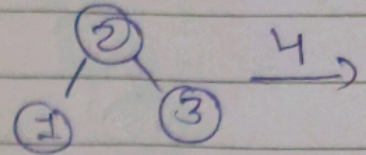


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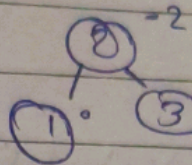
Q.1) 1, 2, 3, 4, 5, 6, 7



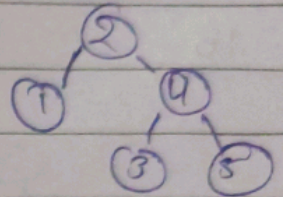
Left Rotation at 1



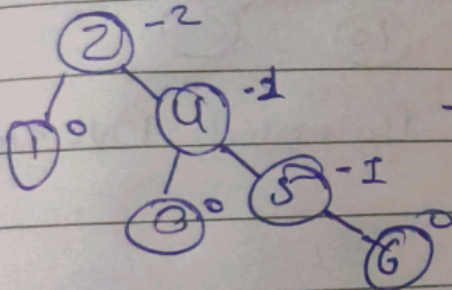
5



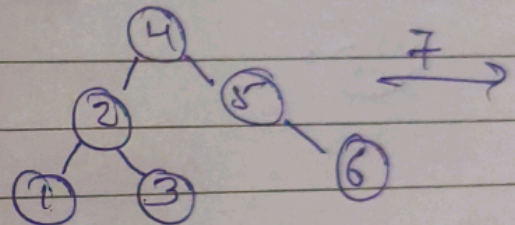
Left at 3



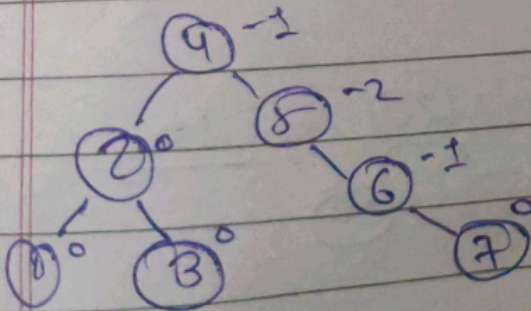
6



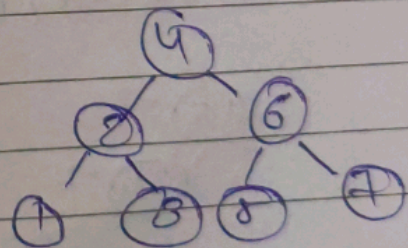
Left at 2



7



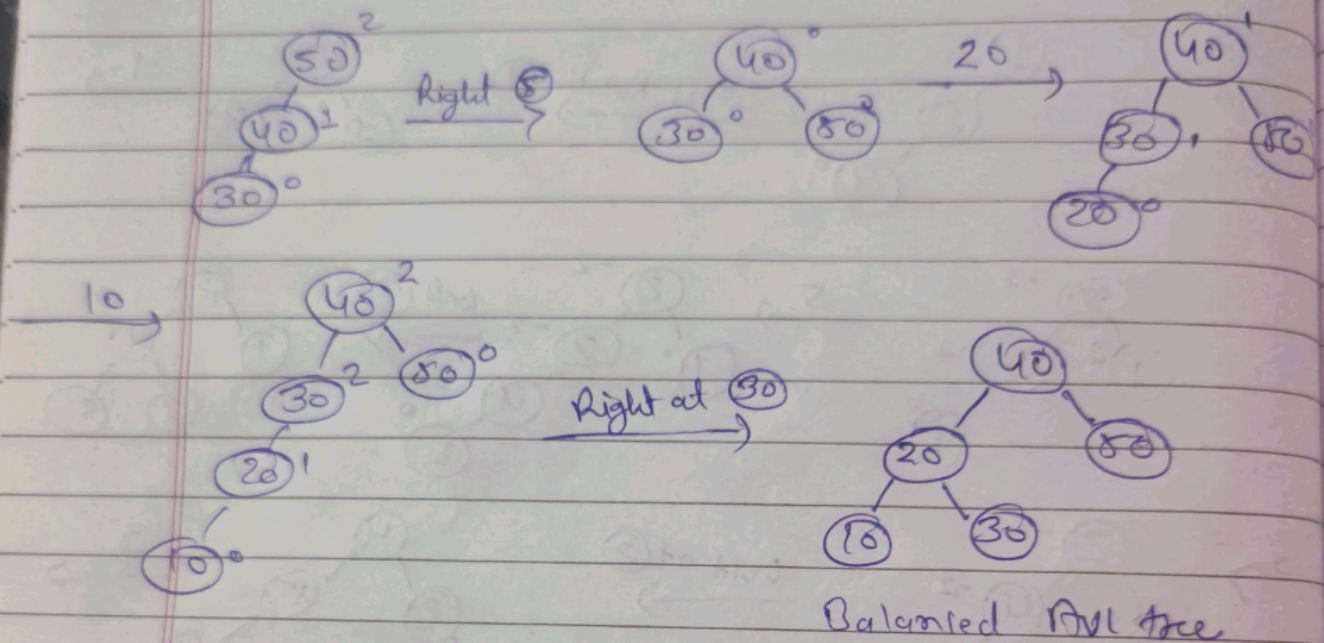
Left at 5



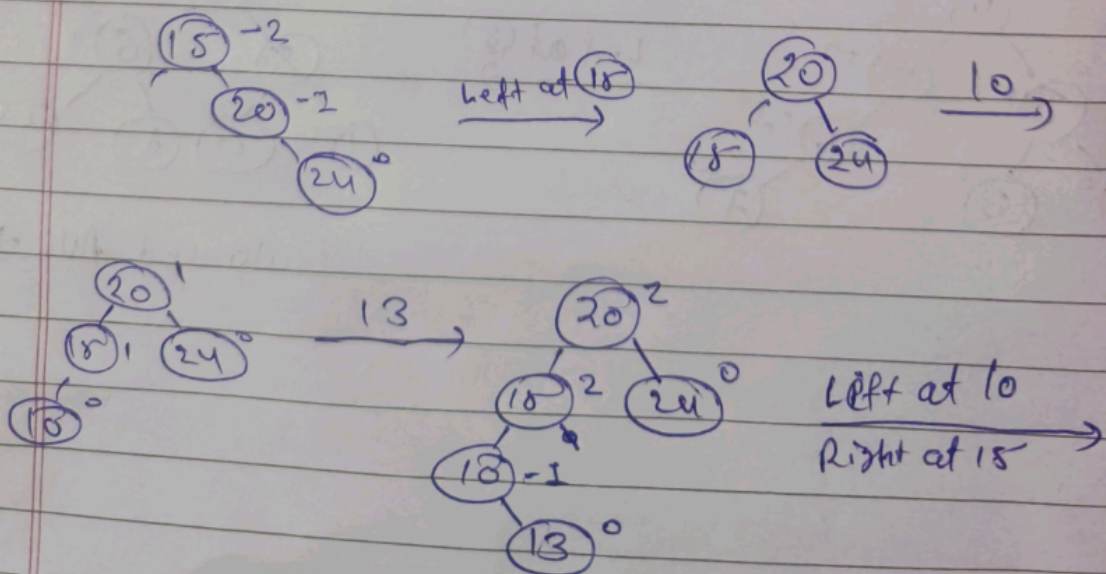
Balanced AVL tree.

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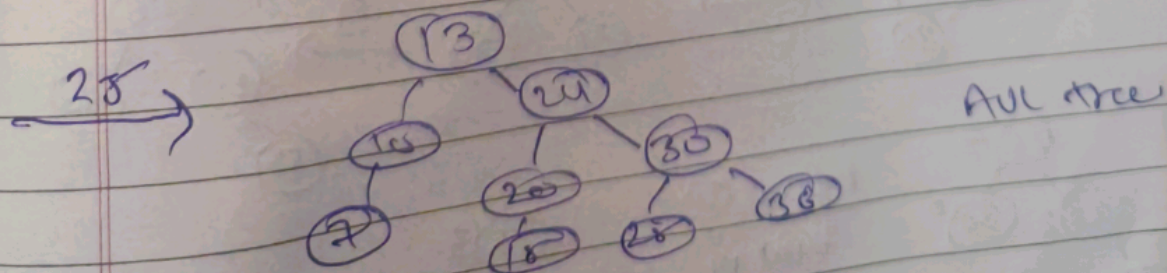
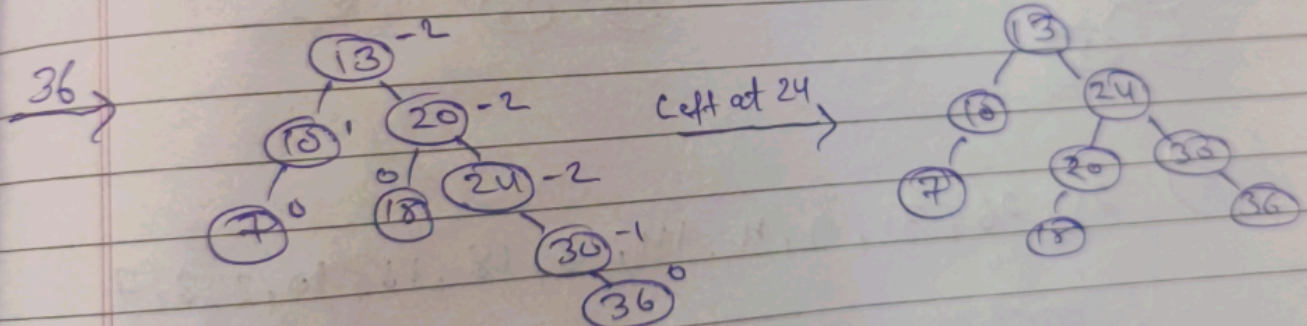
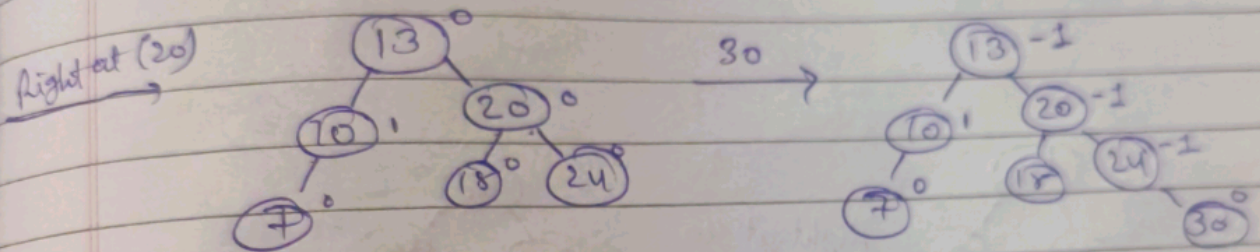
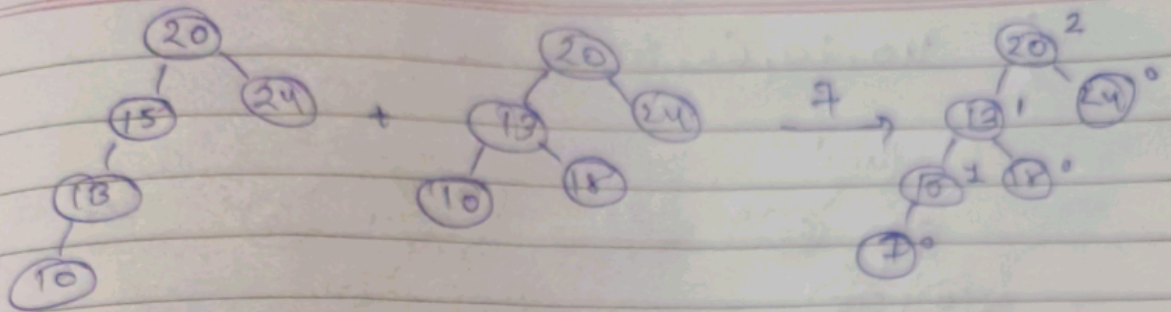
② 50, 40, 30, 20, 10



③ 15, 20, 24, 10, 13, 7, 30, 36, 28



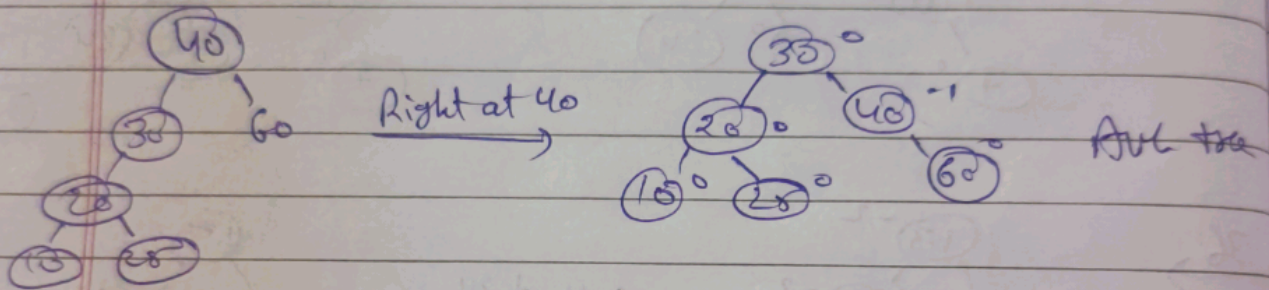
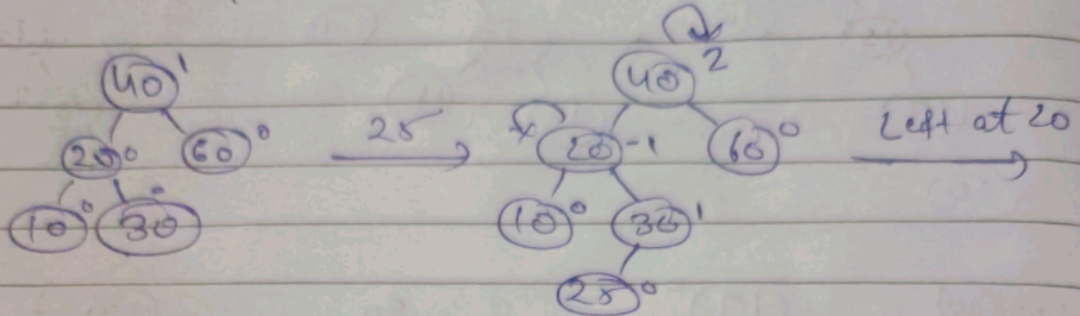
Date: _____



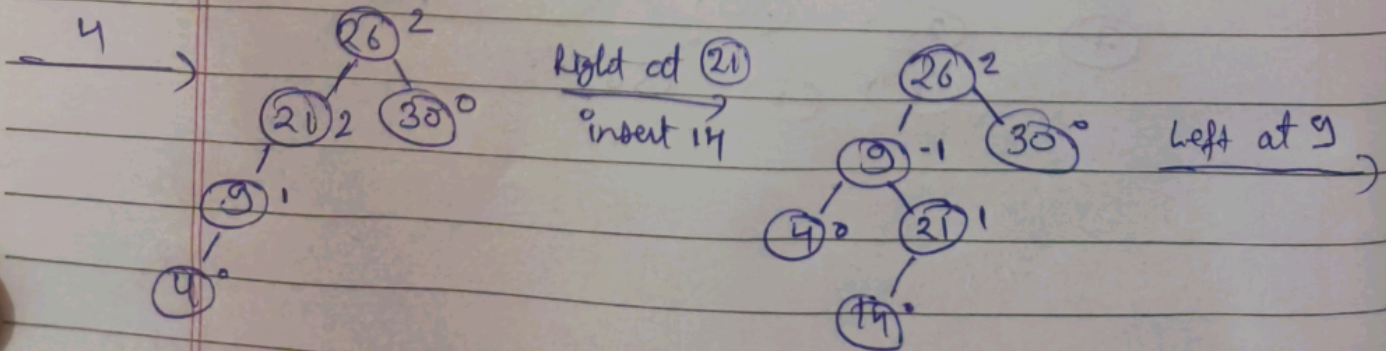
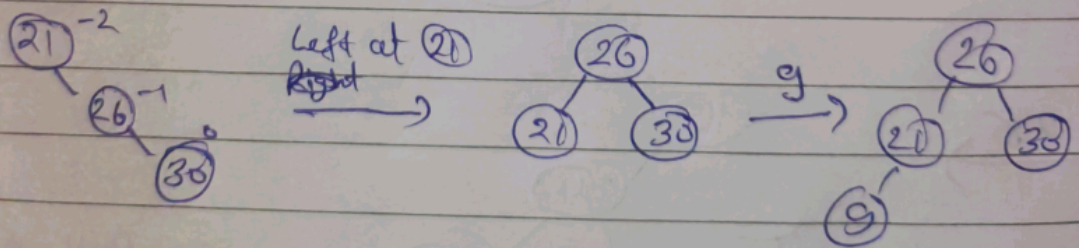
Avl tree

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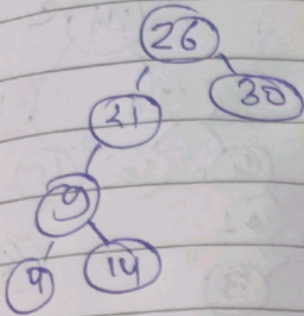
4) 40, 20, 60, 10, 30, 25



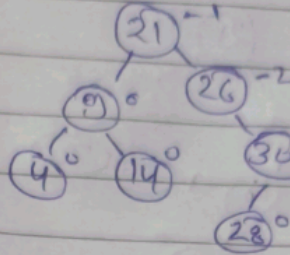
5) 21, 26, 30, 9, 11, 14, 28, 18, 15, 10, 2, 3, 7



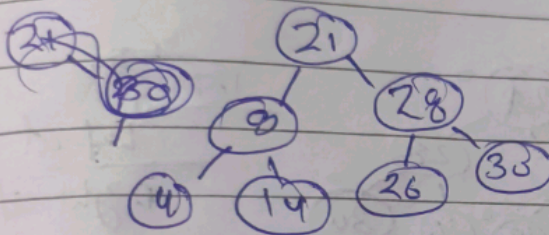
Date.....



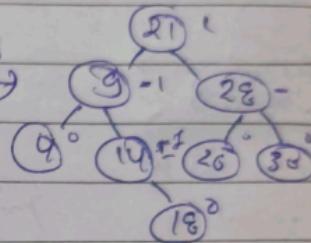
Right at 26
insert 28



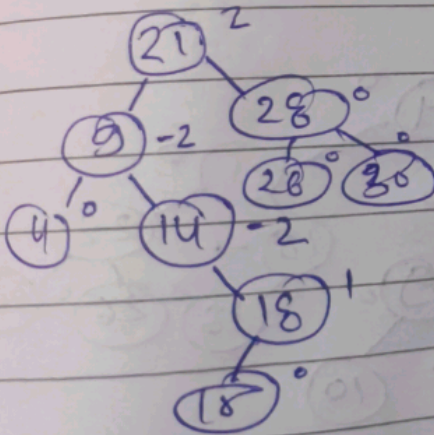
Right at 20
Left at 26
insert 18
Left at 26



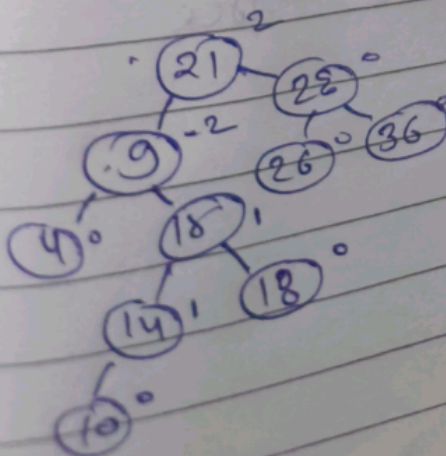
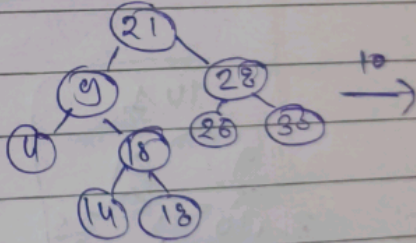
insert 18



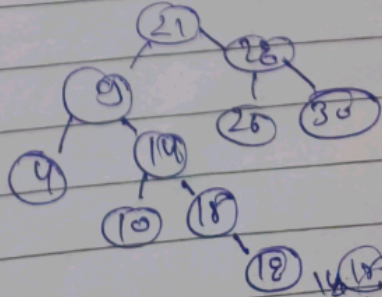
insert 15



Right at 18
Left at 14

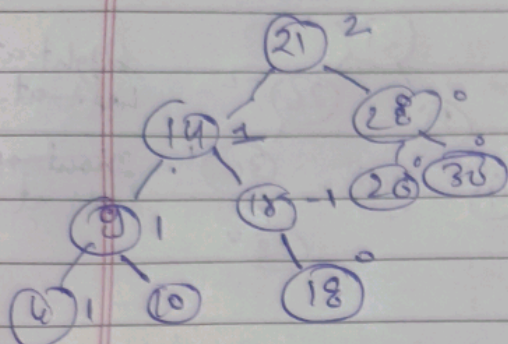


Right at 15

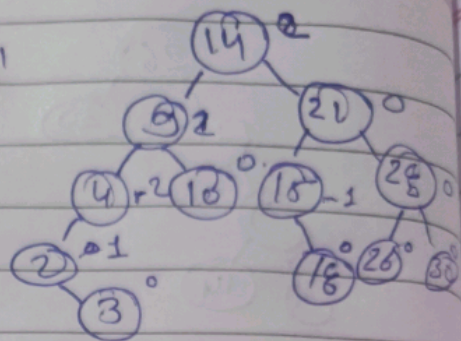


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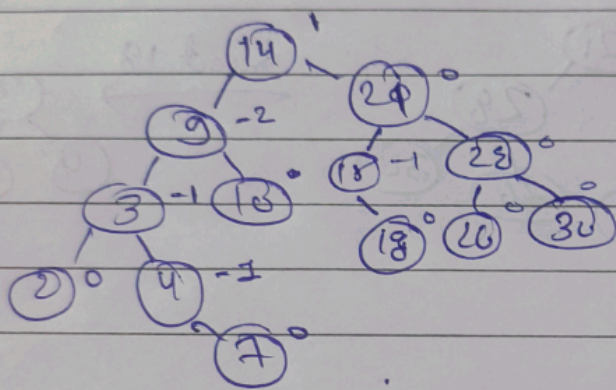
Abhi



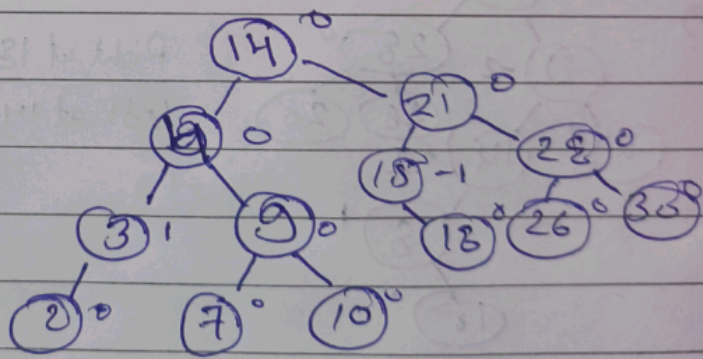
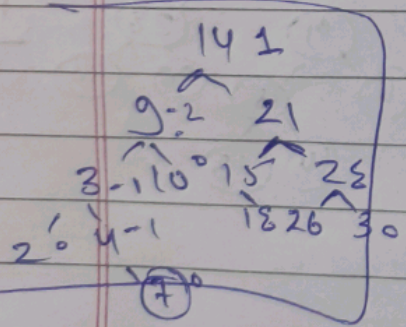
Right at 21
~~insert 2~~
 insert 3



Right at 3
~~insert 3~~
 Left at 4
 insert 7



~~insert 7~~
 Left at 3
 Right at 9



AVL Tree